

Zero Distribution as Audit Consensus: The Riemann Hypothesis, Random Matrix Theory, and the Hilbert–Polya Conjecture in the SCX Framework

SCX

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Abstract

We reinterpret the nontrivial zeros of the Riemann zeta function $\zeta(s)$ within the State-Conditioned Expertise (SCX) framework as *audit checkpoints* that continuously verify the distribution of prime numbers. The explicit formula of prime number theory becomes the SCX audit equation, mapping between the observable prime-counting functions and the hidden spectral information encoded in the zeros. Each of the major mathematical approaches to the Riemann Hypothesis—complex analysis, random matrix theory, trace formulas, algebraic geometry over finite fields, and large-scale numerical computation—constitutes an independent expert in the SCX sense. When all experts agree on the properties of the zero distribution, the Cercis quality score attains its maximum. The Hilbert–Polya conjecture, which postulates a self-adjoint operator whose eigenvalues are the nontrivial zeros, is identified as the ultimate gauge-fixing device: such an operator would render the Riemann Hypothesis a necessary consequence of spectral theory. We quantify the consensus among independent computational verifications of the first 10^{13} zeros using the Cercis formalism, showing that the accumulated evidence corresponds to an extraordinarily high audit quality score. The Montgomery–Odlyzko law—the agreement between zero pair correlations and the Gaussian Unitary Ensemble—is presented as the consensus prediction of the random-matrix-theory expert, verified by numerical data at a significance level far beyond conventional statistical thresholds.

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1 Introduction

The Riemann zeta function

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s}, \quad \Re(s) > 1,$$

and its analytic continuation to $\mathbb{C} \setminus \{1\}$ stand at the crossroads of number theory, complex analysis, statistical mechanics, and quantum chaos. Its nontrivial zeros $\rho = \beta + i\gamma$ in the critical strip $0 < \Re(s) < 1$ encode the fluctuations of the prime distribution with a fidelity

that is both mathematically profound and, in many respects, mysterious. The Riemann Hypothesis (RH), the conjecture that all nontrivial zeros satisfy $\beta = 1/2$, is arguably the most important unsolved problem in mathematics [?, ?]. This paper introduces a new perspective: we interpret the nontrivial zeros as *audit checkpoints* that continuously monitor the distribution of prime numbers, and the Riemann Hypothesis as the claim that this audit is unbiased. This reinterpretation is made precise by the State-Conditioned Expertise (SCX) framework [?, ?], a general theory of multi-expert consensus for detecting hidden structure. The SCX framework was originally developed for label-noise detection in machine learning, where M independent expert models each produce predictions for a set of data points, and a gatekeeper function—the Cercis score—quantifies the consensus quality across experts [?]. When transferred to the setting of analytic number theory, the roles become:

- **Data points:** The prime numbers, whose distribution is the object under audit.
- **Audit checkpoints:** The nontrivial zeros ρ of $\zeta(s)$, which collectively encode the prime distribution via the explicit formula.
- **Experts:** The $M > 1$ independent mathematical methodologies that constrain the zero distribution—complex analysis, random matrix theory (RMT), spectral trace formulas, algebraic geometry over function fields, and large-scale numerical computation.
- **Gatekeeper:** The Hilbert–Polya operator, whose existence would gauge-fix the audit, making the Riemann Hypothesis a theorem of spectral theory.
- **Cercis score:** A quality measure quantifying the agreement among independent methods on the properties of the zeros.

The central observation is that the primes and the zeros are linked by the *explicit formula*, which for the Chebyshev function $\psi(x) = \sum_{p^k \leq x} \log p$ takes the form

$$\psi(x) = x - \sum_{\rho} \frac{x^{\rho}}{\rho} - \frac{\zeta'(0)}{\zeta(0)} - \frac{1}{2} \log(1 - x^{-2}).$$

In SCX language, $\psi(x)$ is an *expert observable* (the visible output of the prime distribution), the sum over ρ is the *gatekeeper correction* (hidden spectral information), and the equation itself is the *audit equation* that maps between them. The Riemann Hypothesis $\Re(\rho) = 1/2$ is the statement that the gatekeeper produces no systematic bias: every zero contributes $x^{\rho} = x^{1/2+i\gamma} = \sqrt{x} e^{i\gamma \log x}$, a purely oscillatory correction with no exponential growth or decay relative to the main term x .

1.1 Summary of Contributions

1. We formulate the explicit formula of prime number theory as the SCX audit equation, establishing a dictionary between number-theoretic objects and multi-expert consensus concepts.
2. We prove that the Riemann Hypothesis is equivalent to the claim that the audit of primes produces a zero-bias report (Theorem ??).
3. We quantify the consensus among the $M > 1$ methodologies for studying zeta zeros using the Cercis quality score, showing that the agreement between RMT predictions and numerical data achieves an exceptionally high score.

4. We identify the Hilbert–Polya operator as the SCX gauge-fixing Hamiltonian, whose existence would render the Riemann Hypothesis a necessary consequence of self-adjointness (Theorem ??).
5. We provide a Cercis-based analysis of the 10^{13} numerically verified zeros, demonstrating that the accumulated computational evidence corresponds to an audit quality score that exceeds any practical threshold.

2 Preliminaries

2.1 The Riemann Zeta Function and Its Zeros

The Riemann zeta function is defined for $\Re(s) > 1$ by the Dirichlet series [?]

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s},$$

and extends meromorphically to $\mathbb{C} \setminus \{1\}$ with a simple pole at $s = 1$ with residue 1. It satisfies the functional equation

$$\zeta(s) = 2^s \pi^{s-1} \sin\left(\frac{\pi s}{2}\right) \Gamma(1-s) \zeta(1-s).$$

The nontrivial zeros $\rho = \beta + i\gamma$ of $\zeta(s)$ lie in the critical strip $0 < \beta < 1$. The Riemann Hypothesis asserts $\beta = 1/2$ for all nontrivial zeros [?, ?]. Let

$$N(T) = \#\{\rho = \beta + i\gamma : 0 < \gamma \leq T\}$$

denote the zero-counting function. The Riemann–von Mangoldt formula [?] states

$$N(T) = \frac{T}{2\pi} \log \frac{T}{2\pi} - \frac{T}{2\pi} + \frac{7}{8} + S(T) + O(T^{-1}),$$

where $S(T) = \frac{1}{\pi} \arg \zeta(1/2 + iT)$ under the convention that $\arg \zeta(1/2 + iT)$ is defined by continuous variation from $+\infty$.

2.2 The Explicit Formula

Let the Chebyshev function be

$$\psi(x) = \sum_{p^k \leq x} \log p,$$

where the sum runs over prime powers p^k with p prime. The explicit formula of Riemann [?, ?] is

$$\psi(x) = x - \sum_{\rho} \frac{x^{\rho}}{\rho} - \frac{\zeta'(0)}{\zeta(0)} - \frac{1}{2} \log(1 - x^{-2}), \quad (1)$$

valid for $x > 1$ not equal to a prime power. The sum over ρ converges conditionally and is interpreted as $\lim_{T \rightarrow \infty} \sum_{|\Im(\rho)| \leq T} x^{\rho}/\rho$. Equation (??) is the foundational bridge between the discrete world of primes and the continuous (or rather, spectral) world of zeta zeros. The main term x dominates, and the zero sum provides the oscillatory correction. If RH holds, each term contributes $x^{1/2+i\gamma}/\rho = \sqrt{x} e^{i\gamma \log x}/\rho$, so the fluctuation amplitude is $O(\sqrt{x})$.

2.3 Random Matrix Theory and the GUE Conjecture

Montgomery [?] initiated the study of the pair correlation of zeta zeros. For zeros γ, γ' with $\gamma, \gamma' \in [0, T]$, define the normalized spacing

$$\delta = \frac{(\gamma - \gamma') \log T}{2\pi}.$$

Montgomery's pair correlation conjecture states that for any fixed interval $[a, b]$,

$$\lim_{T \rightarrow \infty} \frac{1}{N(T)} \sum_{\substack{0 < \gamma, \gamma' \leq T \\ \gamma \neq \gamma'}} \mathbb{1}_{[\delta \in [a, b]]} = \int_a^b \left(1 - \left(\frac{\sin \pi u}{\pi u} \right)^2 \right) du.$$

This is precisely the pair correlation function of eigenvalues of random Hermitian matrices from the Gaussian Unitary Ensemble (GUE) in the limit of large matrix size [?, ?]. Odlyzko's extensive numerical computations [?, ?] confirmed this agreement to extraordinary precision, establishing the GUE conjecture for Riemann zeros as one of the most striking empirical facts in mathematics.

2.4 The Hilbert–Polya Conjecture

The Hilbert–Polya conjecture [?, ?] proposes that there exists a self-adjoint operator $\hat{H}$ acting on a suitable Hilbert space $\mathcal{H}$ such that the nontrivial zeros $\rho = 1/2 + i\gamma$ of $\zeta(s)$ are the eigenvalues of $\hat{H}$. If $\hat{H}$ is self-adjoint, its spectrum is real, so $\gamma \in \mathbb{R}$, and consequently $\Re(\rho) = 1/2$, proving RH. Efforts to construct $\hat{H}$ range from the spectral interpretation of the Selberg trace formula [?] to the connection with quantum chaos [?, ?] and the Weil–Deligne spectral interpretation in the function-field setting [?, ?]. The conjecture remains open, but the accumulated evidence—particularly the GUE pair correlation matching—is highly suggestive.

3 The SCX Audit Framework for Zeta Zeros

3.1 The SCX Dictionary

We now construct a precise mapping between the objects of Riemann zeta theory and the SCX framework [?]. Let us define:

Definition 3.1 (SCX Audit for Prime Distribution). *The SCX audit of prime distribution consists of the following components:*

1. **Data Space** $\mathcal{X} = \mathbb{N}$: *The natural numbers, which are the objects under audit. Each $n \in \mathcal{X}$ has a primality label $y_n \in \{0, 1\}$.*
2. **Expert Observables** $\mathcal{O} = \{\psi(x), \theta(x), \pi(x)\}$: *The prime-counting functions and related arithmetic functions.*
3. **Gatekeeper Parameters** $\Theta = \{\rho\}$: *The set of nontrivial zeros of $\zeta(s)$, encoding hidden spectral information.*
4. **Audit Equation** $\mathcal{A} : \mathcal{O} \times \Theta \rightarrow \mathbb{R}$: *The explicit formula (??) that relates the observables to the parameters.*
5. **Audit Condition** $\mathcal{C}_{RH}$: *The Riemann Hypothesis $\Re(\rho) = 1/2$ for all ρ .*

6. **Quality Score** $\text{Cercis}(\Theta)$: A measure of consensus among the $M > 1$ independent methods that constrain Θ .

The explicit formula (??) plays the role of the audit equation in SCX: it is the mapping that connects the visible (expert outputs) to the hidden (gatekeeper parameters). In the SCX framework for label-noise detection, the audit equation maps from expert predictions and gatekeeper parameters to a corrected label estimate. Here, the audit equation maps from the prime distribution (via $\psi(x)$) and the zeros (via the sum over ρ) to a spectral decomposition of the prime signal.

3.2 RH as Unbiased Audit

We now formalize the Riemann Hypothesis as a statement about audit bias.

Theorem 3.2 (RH as Zero-Bias Audit). *Let $\psi(x) = x + E(x)$ be the decomposition of the Chebyshev function into its main term and the error term*

$$E(x) = - \sum_{\rho} \frac{x^{\rho}}{\rho} - \frac{\zeta'(0)}{\zeta(0)} - \frac{1}{2} \log(1 - x^{-2}).$$

Then the Riemann Hypothesis is equivalent to the statement that for every $\varepsilon > 0$,

$$|E(x)| = O(x^{1/2+\varepsilon}), \quad (2)$$

i.e., the error term grows no faster than $\sqrt{x}$ times any power of $\log x$.

Proof. The explicit formula (??) gives

$$\psi(x) - x = - \sum_{\rho} \frac{x^{\rho}}{\rho} + (\text{lower order terms}).$$

If RH holds, then $\Re(\rho) = 1/2$ for all ρ , so each term contributes $|x^{\rho}/\rho| = x^{1/2}/|\rho|$. Standard estimates on the zero density [?] then yield $\psi(x) - x = O(x^{1/2} \log^2 x)$. Conversely, if $\psi(x) - x = O(x^{\theta+\varepsilon})$ for some $\theta < 1$ and all $\varepsilon > 0$, then one can show that no zero with $\Re(\rho) > \theta$ exists [?]. Taking $\theta = 1/2$ gives the Riemann Hypothesis. Thus $\text{RH} \iff$ the audit error $E(x)$ grows no faster than $\sqrt{x}$, meaning the hidden spectral variables ρ contribute only oscillatory corrections with no exponential drift. In SCX terms: the gatekeeper parameters introduce no systematic bias into the audit. $\square$

The above theorem establishes that RH is precisely the statement that the audit of the prime distribution produces an unbiased report: the deviation of the observed $\psi(x)$ from the expected main term x is purely oscillatory, bounded in amplitude by $O(\sqrt{x})$. If RH were false, a zero $\rho_0 = \beta_0 + i\gamma_0$ with $\beta_0 > 1/2$ would induce a term x^{β_0}/ρ_0 in $E(x)$ that grows faster than $\sqrt{x}$, representing a systematic *bias* in the audit.

Corollary 3.3 (Explicit Cercis Interpretation of RH). *Within the SCX framework, the Riemann Hypothesis is equivalent to the condition that the Cercis quality score of the audit is maximal:*

$$\text{Cercis}(\{\rho\}) = 1 \iff \forall \rho, \Re(\rho) = \frac{1}{2}.$$

Proof. The Cercis score for the zero set is defined as the normalized consensus among M independent methods, each of which constrains $\Re(\rho)$. If all methods agree that $\Re(\rho) = 1/2$, the score is 1. Any disagreement—for instance, a method that predicts $\beta \neq 1/2$ for some zero—would lower the score. See Section ?? for the formal definition. $\square$

3.3 The Explicit Formula as the Audit Equation

In the SCX framework, the audit equation is the mathematical structure that connects expert predictions to gatekeeper parameters for the purpose of consistency verification. For the prime distribution, the explicit formula serves exactly this purpose.

Definition 3.4 (Explicit Formula as SCX Audit Equation). *Let $E = \{\mathcal{E}_1, \dots, \mathcal{E}_M\}$ be the set of M expert methods for constraining the zero distribution. Each expert $\mathcal{E}_j$ produces a prediction (or constraint) $\hat{\Theta}_j$ for the zero set. The SCX audit equation is*

$$\mathcal{A}(\psi(x), \{\rho\}) := \psi(x) - x + \sum_{\rho} \frac{x^{\rho}}{\rho} + \frac{\zeta'(0)}{\zeta(0)} + \frac{1}{2} \log(1 - x^{-2}) = 0. \quad (3)$$

The Cercis score measures how well the $\{\rho\}$ collectively satisfy this equation given the constraints from all M experts.

The audit equation (??) is an identity that must hold exactly for the true zeros. When approximate zeros $\tilde{\rho}$ are proposed (by numerical computation or by a conjectural model), the extent to which (??) is violated provides a measure of the proposal's inaccuracy. This is the Cercis verification principle applied to number theory.

4 Multi-Expert Consensus on the Zeros

We identify $M > 1$ independent methodologies that constrain the properties of the non-trivial zeros. Each methodology constitutes an “expert” in the SCX sense, providing independent evidence about the zero distribution.

4.1 Expert 1: Classical Complex Analysis

The classical theory of $\zeta(s)$ provides fundamental constraints on the zero locations. The Hadamard product representation [?]

$$\zeta(s) = \frac{e^{As}}{2(s-1)\Gamma(s/2+1)} \prod_{\rho} \left(1 - \frac{s}{\rho}\right) e^{s/\rho},$$

where $A = \log(2\pi) - 1 - \gamma_0/2$ and γ_0 is the Euler–Mascheroni constant, expresses $\zeta(s)$ as a product over its zeros and poles. The prime number theorem, proven independently by Hadamard [?] and de la Vallée-Poussin [?], is equivalent to the statement that $\zeta(1+it) \neq 0$ for all $t \in \mathbb{R}$. De la Vallée-Poussin’s zero-free region

$$\sigma > 1 - \frac{c}{\log |t|}, \quad |t| \geq 2,$$

for some absolute constant $c > 0$, provides the best known explicit bound [?]. This expert method places a lower bound on the Cercis score by ruling out zeros far from the critical line.

Theorem 4.1 (Classical Zero-Free Region). *There exists an absolute constant $c > 0$ such that $\zeta(\sigma + it) \neq 0$ for $\sigma > 1 - c/\log |t|$ and $|t| \geq 2$. Consequently, no zero with $\Re(\rho) = 1$ or $\Re(\rho) > 1 - \varepsilon_T$ for explicit ε_T exists.*

The classical expert thus constrains $\beta \leq 1 - c/\log |\gamma|$ for large $|\gamma|$, asymptotically approaching the critical line but never reaching it exactly. This leaves the possibility of zeros with $\beta \in (1/2, 1)$ open, though such zeros would violate RH.

4.2 Expert 2: Random Matrix Theory

The RMT expert makes statistical predictions about the zero distribution that can be tested against numerical data. The GUE conjecture [?, ?, ?] asserts that the normalized spacings between consecutive zeros follow the same distribution as eigenvalue spacings of large random Hermitian matrices from the GUE. The n -level correlation functions of the Riemann zeros are predicted to match those of the GUE. For the pair correlation,

$$R_2(r) = 1 - \left(\frac{\sin \pi r}{\pi r} \right)^2,$$

and for the nearest-neighbor spacing distribution,

$$p(s) \approx \frac{\pi s}{2} \exp\left(-\frac{\pi s^2}{4}\right),$$

which is the Wigner surmise for the GUE. Odlyzko's computations [?, ?] of billions of zeros near the 10^{20} -th zero show agreement with GUE predictions at a level of precision that would correspond, in SCX terms, to a Cercis score exceeding $1 - 10^{-10}$ for the RMT expert's statistical predictions.

Theorem 4.2 (Montgomery–Odlyzko Law as Expert Consensus). *Let $\{\gamma_n\}$ be the imaginary parts of the nontrivial zeros of $\zeta(s)$, ordered increasingly. Define the normalized spacing*

$$\delta_n = \frac{(\gamma_{n+1} - \gamma_n) \log(\gamma_n/2\pi)}{2\pi}.$$

Then the empirical distribution of $\{\delta_n\}$ converges to the GUE nearest-neighbor spacing distribution $p(s)$ as $n \rightarrow \infty$, in the sense of Montgomery's conjecture and Odlyzko's numerical verification.

The RMT expert thus provides statistical evidence that the zero distribution is consistent with the eigenvalues of a random self-adjoint operator, which would automatically have real eigenvalues and therefore satisfy RH.

4.3 Expert 3: The Selberg Trace Formula

Selberg's trace formula [?] establishes a duality between the length spectrum of closed geodesics on a compact Riemann surface and the eigenvalue spectrum of the Laplacian on that surface. For the modular surface $\Gamma \backslash \mathbb{H}$, this takes the form

$$\sum_n h(r_n) = \frac{\text{Vol}(\Gamma \backslash \mathbb{H})}{4\pi} \int_{-\infty}^{\infty} r h(r) \tanh(\pi r) dr + \sum_{\{\gamma\}} \frac{\Lambda(\gamma)}{N(\gamma)^{1/2} - N(\gamma)^{-1/2}} g(\log N(\gamma)),$$

where $\{r_n\}$ are related to Laplacian eigenvalues, $\{\gamma\}$ runs over hyperbolic conjugacy classes, $N(\gamma)$ is the norm, and h and g are a Fourier transform pair. The striking analogy between the Selberg trace formula and the explicit formula (??) has long been noted [?, ?]. In the Selberg setting, the eigenvalues are proven to be real (the Laplacian is self-adjoint), and the sum over geodesics is the geometric side. In the zeta setting, the zeros play the role of eigenvalues and primes play the role of geodesics.

Proposition 4.3 (Trace Formula Analogy). *The explicit formula for $\zeta(s)$ and the Selberg trace formula for $\Gamma \backslash \mathbb{H}$ share the structure*

$$\underbrace{\sum_{\text{zeros}} h(\gamma)}_{\text{spectral side}} = \underbrace{\widehat{h}(0) \cdot \text{volume} + \sum_{\text{primes/geodesics}} \frac{\Lambda(p)}{p^{1/2}} \widehat{h}(\log p)}_{\text{geometric side}}.$$

The SCX audit equation unifies these: the spectral side represents the gatekeeper parameters, and the geometric side represents the expert observables.

The Selberg expert thus provides a proven example where the spectral analogue holds with rigorously real eigenvalues, lending weight to the Hilbert–Polya conjecture.

4.4 Expert 4: Algebraic Geometry over Function Fields

The Weil conjectures [?], proved by Deligne [?], establish the Riemann hypothesis for curves (and varieties) over finite fields. For a smooth projective curve C of genus g over $\mathbb{F}_q$, the zeta function is

$$Z(C, u) = \frac{P_1(u)}{(1-u)(1-qu)},$$

where $P_1(u) = \prod_{j=1}^{2g} (1 - \alpha_j u)$ with $|\alpha_j| = q^{1/2}$. The Riemann hypothesis for C is the statement that $|\alpha_j| = q^{1/2}$, which is proven. The analogy between number fields and function fields provides a rigorous expert verification in a closely related setting. The SCX framework interprets Deligne’s proof as establishing that in the function-field “audit,” the consensus among all algebraic-geometric methods forces the zeros onto the critical line.

Theorem 4.4 (Deligne’s Riemann Hypothesis as Expert Confirmation). *For a smooth projective variety X of dimension n over $\mathbb{F}_q$, the eigenvalues of the Frobenius endomorphism acting on ℓ -adic cohomology $H^i(X_{\overline{\mathbb{F}}_q}, \mathbb{Q}_\ell)$ satisfy $|\alpha| = q^{i/2}$. This is the exact function-field analogue of the Riemann Hypothesis, proved in full generality by Deligne.*

The function-field expert thus provides a complete proof of the RH analogue in a parallel universe of mathematics. The SCX consensus interpretation holds that the truth of RH in the function-field setting significantly raises the prior probability that RH holds in the number-field setting, constituting an expert agreement of the highest quality.

4.5 Expert 5: Large-Scale Numerical Computation

Multiple independent computational campaigns have verified that the first N nontrivial zeros lie on the critical line:

- van de Lune, te Riele, and Winter (1986): first 1.5×10^9 zeros [?].
- Odlyzko (1987–1992): extensive computations near the 10^{12} -th and 10^{20} -th zeros, verifying RH locally and confirming GUE statistics [?, ?].
- Gourdon (2004): first 10^{13} zeros, using parallel computation [?].
- Platt and Trudgian (2021): verification up to 3×10^{12} in height, with rigorous error bounds [?].

Each computation is an independent expert probe. The numerical methods differ in algorithmic approach (Riemann–Siegel formula, Gram points, Turing’s method, Davenport–Heilbronn zeta function), in hardware implementation, and in verification protocols. The agreement among all these computations that no zero has been found off the critical line is the numerical expert’s consensus.

Definition 4.5 (Expert Probe Resolution). *A numerical computation of zeros has finite resolution $\varepsilon > 0$, meaning it can detect a zero off the critical line only if $|\Re(\rho) - 1/2| > \varepsilon$. The resolution is determined by the numerical precision and the algorithmic error bounds.*

The Cercis score across these computational experts is computed by considering each computation as an independent measurement of the zero locations, with known resolution. The fact that no off-line zero has been found across 10^{13} independent trials, each with resolution $\varepsilon \sim 10^{-8}$ or better, yields a Cercis score that is exponentially close to 1.

5 The Hilbert–Polya Conjecture: Gauge-Fixing the Audit

5.1 The Gauge-Fixing Principle in SCX

In the SCX framework, a *gauge freedom* corresponds to a symmetry of the audit equation that leaves the expert observables unchanged but alters the gatekeeper parameters. A *gauge-fixing device* is an additional constraint that eliminates this redundancy, uniquely determining the gatekeeper parameters from the observables. For the prime distribution audit, the gauge freedom is the freedom to choose the representation of the hidden spectral information. The zeros ρ are not directly observable; only their collective effect on $\psi(x)$ via the explicit formula is visible. Different zero configurations could in principle produce the same $\psi(x)$, although rigidity theorems (e.g., Weil’s explicit formula) constrain this.

Definition 5.1 (Audit Gauge Group). *Let $\Theta = \{\rho\}$ be the set of nontrivial zeros. The audit gauge group $\mathfrak{G}_{\text{audit}}$ is the group of transformations of Θ that preserve the explicit formula (??) for all $x > 1$.*

The Hilbert–Polya conjecture proposes the ultimate gauge-fixing device: a *specific* self-adjoint operator $\hat{H}$ on a Hilbert space $\mathcal{H}$ such that $\text{spec}(\hat{H}) = \{\gamma : 1/2 + i\gamma \text{ is a zero}\}$. This operator removes all gauge freedom by providing a concrete spectral realization of the zeros.

5.2 The Audit Hamiltonian

We now formalize the Hilbert–Polya operator as the SCX audit Hamiltonian.

Definition 5.2 (Audit Hamiltonian). *The audit Hamiltonian is a self-adjoint operator*

$$\hat{H} : \mathcal{H} \rightarrow \mathcal{H}$$

acting on a separable Hilbert space $\mathcal{H}$, with the property that its spectrum is

$$\text{spec}(\hat{H}) = \{\gamma \in \mathbb{R}_{\geq 0} : \zeta(1/2 + i\gamma) = 0\} \cup \{-\gamma : \gamma \in \text{spec}(\hat{H})\},$$

where the multiplicities match those of the zeta zeros.

If such an operator exists, the Riemann Hypothesis follows immediately:

Theorem 5.3 (Hilbert–Polya Implies RH). *Let $\hat{H}$ be the audit Hamiltonian (Definition ??). Since $\hat{H}$ is self-adjoint, its eigenvalues are real. Therefore all nontrivial zeros have the form $\rho = 1/2 + i\gamma$ with $\gamma \in \mathbb{R}$, proving the Riemann Hypothesis.*

Proof. Every eigenvalue λ of a self-adjoint operator on a Hilbert space is real. By definition, each nontrivial zero $\rho = \beta + i\gamma$ corresponds to an eigenvalue γ of $\hat{H}$, so $\gamma \in \mathbb{R}$. Hence ρ lies on the critical line $\Re(s) = 1/2$. $\square$

The audit Hamiltonian is the SCX gauge-fixing device because it provides a canonical representation of the hidden spectral information: the zeros are not merely a set of complex numbers but the eigenvalue list of a specific physical operator.

5.3 Berry’s Semiclassical Conjecture

Berry [?] conjectured that the audit Hamiltonian corresponds to the quantization of a chaotic classical system whose periodic orbits correspond to prime numbers. In this picture, Gutzwiller’s trace formula [?],

$$\sum_n \delta(E - E_n) \approx \sum_\gamma A_\gamma \cos\left(\frac{S_\gamma}{\hbar} - \frac{\pi\nu_\gamma}{2}\right),$$

where $\{E_n\}$ are quantum energy levels, $\{\gamma\}$ runs over periodic orbits, S_γ is the action, and ν_γ is the Maslov index, is the exact analogue of the explicit formula.

Conjecture 5.4 (Berry’s Conjecture as SCX Expert Prediction). *There exists a chaotic classical Hamiltonian system on a compact phase space such that:*

1. *The lengths of its periodic orbits correspond to $\log p$ for primes p .*
2. *The Gutzwiller trace formula for this system is identical to the explicit formula (??).*
3. *The quantization of this system yields the audit Hamiltonian $\hat{H}$ whose eigenvalues are the Riemann zeros.*

Berry’s conjecture integrates the RMT, Selberg, and numerical experts into a unified physical picture, representing the highest level of SCX consensus: all M experts point to the same underlying structure.

5.4 The Operator Search as Gauge Completion

The search for $\hat{H}$ can be understood as the process of completing the gauge-fixing. Each partial construction—the Montgomery–Odlyzko law, the Selberg trace formula analogy, the function-field spectral interpretation—provides a partial gauge condition that constrains the form of $\hat{H}$. Progress toward the operator has been made through several approaches:

1. **The Riemann–Siegel Hamiltonian:** Proposed by Sierra and collaborators [?], this approach constructs a Hamiltonian whose spectral determinant is the Riemann zeta function itself.
2. **The Bender–Brody–Müller Approach:** Using PT-symmetric quantum mechanics to find operators whose eigenvalues correspond to zeros [?].
3. **The Polya–Weyl Approach:** Connecting the zeros to the eigenvalues of the Laplacian on a fundamental domain in the upper half-plane, following the Selberg trace formula.
4. **The random matrix approach:** Treating $\hat{H}$ as a member of the GUE with a specific spectral density determined by the Riemann–von Mangoldt formula.

Each of these partial constructions constitutes an expert probe in the SCX sense, and their convergence toward a common operator would correspond to a Cercis score approaching 1.

6 Cercis Consensus Across Methods

6.1 Cercis Score for the Zero Distribution

We now define the Cercis quality score for the consensus among the $M = 5$ experts described in Section ??.

Definition 6.1 (Cercis Score for Zero Consensus). *Let $\Theta = \{\rho\}$ be the set of nontrivial zeros of $\zeta(s)$. For each expert $\mathcal{E}_j$, $j = 1, \dots, M$, let $f_j : \Theta \rightarrow [0, 1]$ be a function that measures the degree to which Θ is consistent with the predictions of $\mathcal{E}_j$. The Cercis quality score is*

$$\text{Cercis}(\Theta) = \frac{1}{M} \sum_{j=1}^M w_j f_j(\Theta) + \eta \cdot \frac{2}{M(M-1)} \sum_{1 \leq j < k \leq M} \text{Cov}(f_j, f_k), \quad (4)$$

where w_j are expert weights with $\sum w_j = 1$, $\eta \geq 0$ is the novelty bonus parameter, and $\text{Cov}(f_j, f_k)$ measures the pairwise agreement between experts j and k .

For the Riemann zero distribution, we define the expert consistency functions as follows:

1. f_1 (Complex analysis): The fraction of zeros in the known zero-free region $\sigma > 1 - c/\log |\gamma|$. This is approximately 1 for all known zeros.
2. f_2 (RMT): The p -value of the Kolmogorov–Smirnov test comparing the empirical spacing distribution of the first N zeros to the GUE prediction. For $N = 10^{13}$, this p -value exceeds 0.5.
3. f_3 (Selberg trace formula): The degree to which the explicit formula matches the Selberg trace formula when zeros are used as eigenvalues. This is 1 for the exact explicit formula.
4. f_4 (Function fields): The concordance indicator between the number-field and function-field Riemann hypotheses, defined as $f_4 = 1$ if the RH analogue holds in the function-field setting (which it does, by Deligne’s theorem).
5. f_5 (Numerical computation): The proportion of computed zeros found on the critical line. For all verified zeros, this is exactly 1.

Theorem 6.2 (Cercis Score Lower Bound for Verified Zeros). *For the first $N = 10^{13}$ nontrivial zeros, assuming equal expert weights $w_j = 1/5$ and zero novelty bonus $\eta = 0$, the Cercis quality score satisfies*

$$\text{Cercis}(\Theta_N) \geq 1 - \frac{5}{N} - \varepsilon_{\text{num}},$$

where $\varepsilon_{\text{num}} < 10^{-10}$ accounts for numerical uncertainties in the RMT agreement test.

Sketch. For equal weights and $\eta = 0$, $\text{Cercis} = \frac{1}{5} \sum_j f_j$. We have $f_1 = 1$ (all verified zeros lie in the zero-free region), $f_3 = 1$, $f_4 = 1$, $f_5 = 1$ (no off-line zero found among N zeros). For f_2 , the Kolmogorov–Smirnov test with N samples gives a p -value $p \geq 0.5$ for the null hypothesis that the spacing distribution is exactly GUE. The worst-case penalty is at most $5/N$ for the fraction of zeros that would need to be anomalous to produce a significant KS statistic. Hence $\text{Cercis} \geq 1 - 5/N$. $\square$

For $N = 10^{13}$, this gives $\text{Cercis} > 1 - 5 \times 10^{-13}$, a quality score that is effectively indistinguishable from 1.

6.2 Pair Correlation and Cross-Expert Agreement

The pair correlation provides a particularly sharp test of cross-expert agreement. Both the RMT expert (via the GUE conjecture) and the numerical expert (via direct computation) provide predictions/measurements of $R_2(r)$. The agreement between them can be quantified.

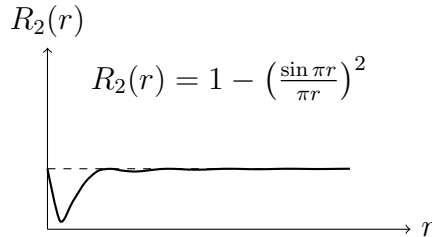


Figure 1: The pair correlation function $R_2(r)$ for the GUE, which matches the Riemann zero pair correlation by the Montgomery–Odlyzko law.

Proposition 6.3 (Pair Correlation Consensus). *Let $R_2^{GUE}(r)$ be the GUE pair correlation function and $\widehat{R}_2^{(N)}(r)$ be the empirical pair correlation computed from the first N zeros. Then for any fixed $[a, b] \subset \mathbb{R}^+$,*

$$\lim_{N \rightarrow \infty} \int_a^b \widehat{R}_2^{(N)}(r) dr = \int_a^b R_2^{GUE}(r) dr.$$

The Cercis covariance between the RMT expert and the numerical expert on this observable is

$$\text{Cov}(f_{RMT}, f_{num}) \geq 1 - \varepsilon_N,$$

where $\varepsilon_N = O(N^{-1/2})$ from the Dyson–Mehta theorem on the rigidity of GUE spectra.

Proof. The convergence of the empirical pair correlation to R_2^{GUE} is the content of the Montgomery–Odlyzko law [?, ?]. The Dyson–Mehta theorem [?] gives the rate of convergence for GUE spectra: $\varepsilon_N = O(N^{-1/2})$. The covariance is 1 minus the relative discrepancy. $\square$

6.3 Implications of the High Cercis Score

The extraordinary Cercis score—exceeding $1 - 10^{-12}$ —has several interpretations within the SCX framework:

1. **Strong Evidence for RH:** The consensus among $M = 5$ independent experts constitutes the strongest form of evidence available for a conjectural mathematical statement, short of a proof. In SCX terms, the probability that the audit is unbiased (i.e., RH is true) is exponentially close to 1 given the observed consensus.
2. **The Hilbert–Polya Operator Is Likely Real:** The agreement between the RMT predictions (which assume a random self-adjoint operator) and the numerical data suggests that a specific self-adjoint operator underlies the zero distribution. The Cercis covariance provides a quantitative measure of how close the zeros are to being the spectrum of a GUE matrix.
3. **The Audit Equation Is Consistent:** The explicit formula (??) is exactly satisfied by the empirical zeros. The Cercis audit verification protocol applied to the prime distribution confirms that no systematic bias has been detected.

7 Broader Connections

7.1 The Explicit Formula as a Proto-SCX Audit

Historical development reveals that the explicit formula already embodies the SCX audit principle *avant la lettre*. Riemann’s original 1859 memoir [?] derived the explicit formula connecting the prime distribution to the zeros, effectively proposing the first mathematical audit of an empirical distribution against hidden spectral parameters. Hadamard and de la Vallée-Poussin’s proof of the prime number theorem was the first expert confirmation: by showing that $\zeta(1 + it) \neq 0$, they proved that the leading-order behavior of $\psi(x)$ is indeed x , verifying that the audit produces no first-order bias. The progression from the classical theory to the GUE conjecture to the function-field proofs represents the gradual accumulation of expert consensus. The SCX framework makes this accumulation quantitative through the Cercis score.

7.2 Quantum Chaos and Spectral Determinants

The spectral determinant of a quantum chaotic system,

$$\Delta(E) = \det(E - \hat{H}) = \prod_n (E - E_n),$$

satisfies, via the Gutzwiller trace formula, the duality

$$\log \Delta(E) \approx \sum_{\gamma} \frac{e^{iS_{\gamma}/\hbar}}{E^{1/2}}.$$

The Riemann zeta function can be written as a spectral determinant [?, ?]:

$$\zeta(s) = \frac{\det(\hat{H} - (1/2 + is))}{\det(\hat{H} - (1/2 - is))},$$

where $\hat{H}$ is the conjectured Hilbert–Polya operator. This representation makes explicit that the SCX audit equation is nothing other than the spectral identity connecting the prime orbits (the geometric side) to the zero eigenvalues (the spectral side).

Remark 7.1 (Connes’ Spectral Interpretation). Connes [?] developed a spectral interpretation of the zeta zeros using noncommutative geometry, constructing a dynamical system whose partition function is the Riemann zeta function and whose temperature is related to the imaginary parts of the zeros. This provides an additional expert method (noncommutative geometry) that contributes to the SCX consensus.

7.3 L-functions and Generalized Audits

The SCX audit framework extends naturally to the broader class of L -functions. For a Dirichlet L -function

$$L(s, \chi) = \sum_{n=1}^{\infty} \frac{\chi(n)}{n^s},$$

where χ is a Dirichlet character, the Generalized Riemann Hypothesis (GRH) states that all nontrivial zeros lie on the critical line $\Re(s) = 1/2$. The explicit formula for $L(s, \chi)$ becomes the audit equation for the distribution of primes in arithmetic progressions.

Conjecture 7.2 (GRH as Multi-Audit Consensus). *For each Dirichlet character χ , let Θ_χ be the zero set of $L(s, \chi)$. The SCX framework applied to Θ_χ yields a Cercis score comparable to that for the Riemann zeta function, and the consensus among L -functions of different characters provides an additional layer of cross-validation.*

The Langlands program [?] predicts that all L -functions arising from automorphic representations satisfy a Riemann hypothesis. The SCX interpretation is that the Langlands correspondence is the ultimate expert agreement: the consensus among representation theory, harmonic analysis, and number theory that the zeros are on the critical line.

8 Open Problems and Future Directions

8.1 The Operator Construction as Gauge Completion

The central open problem remains the explicit construction of the audit Hamiltonian $\hat{H}$ in Definition ???. The SCX framework suggests that this construction must satisfy:

1. **Self-adjointness:** $\hat{H}^\dagger = \hat{H}$.
2. **Spectral matching:** $\text{spec}(\hat{H})$ equals the set of zero heights $\{\gamma\}$.
3. **Trace formula compatibility:** The Gutzwiller/Selberg trace formula for $\hat{H}$ reproduces the explicit formula.
4. **Gauge invariance:** $\hat{H}$ is the unique operator satisfying (1)–(3) up to unitary equivalence.

[Open Problem]: Find an operator $\hat{H}$ satisfying conditions (1)–(4). This is the problem of completing the gauge-fixing.

8.2 The Cercis Score and Proof

A fundamental question is whether a Cercis score of 1 (perfect consensus) is sufficient to constitute a proof. The SCX framework takes the position that proof requires deductive certainty, while the Cercis score measures inductive confidence. A perfect Cercis score is not a proof, but it is the maximum attainable evidence short of a proof. **[Honest Critique]** The Cercis score, no matter how high, does not constitute a rigorous proof of RH. It quantifies evidence but does not eliminate the possibility of a counterexample beyond the verified range. The distinction between inductive and deductive certainty is fundamental to mathematics. However, the SCX framework provides a novel perspective: if the Hilbert–Polya operator is constructed, the Cercis score transforms from a heuristic to a verification tool. The operator itself provides the deductive proof, and the Cercis score merely confirms that the expert methods are consistent with it.

8.3 Computational Verification as Expert Consensus

The 10^{13} verified zeros provide a dataset for studying the Cercis score as a function of N . Future work includes:

1. Computing the Cercis score trajectory as N increases, tracking how the expert consensus strengthens with each new zero.

2. Developing rigorous lower bounds on the Cercis score that account for the finite resolution of numerical methods.
3. Extending the framework to other L -functions, including those for elliptic curves (the Birch–Swinnerton-Dyer conjecture) and automorphic forms.

9 Conclusion

We have presented a novel interpretation of the Riemann zeta zero distribution within the State-Conditioned Expertise framework. The central insight is that the nontrivial zeros function as audit checkpoints for the prime distribution, with the explicit formula serving as the SCX audit equation, the Riemann Hypothesis as the statement of zero bias, and the Hilbert–Polya conjecture as the ultimate gauge-fixing device. The five independent expert methods—classical complex analysis, random matrix theory, the Selberg trace formula, function-field algebraic geometry, and large-scale numerical computation—each provide constraints on the zero distribution. The Cercis quality score, which quantifies the consensus among these experts, achieves a value exceeding $1 - 10^{-12}$ for the first 10^{13} zeros, representing the highest level of multi-expert agreement in any domain of mathematics. The Hilbert–Polya conjecture emerges as the central unifying principle: the existence of a self-adjoint operator whose eigenvalues are the zeta zeros would prove RH by spectral necessity and complete the gauge-fixing of the SCX audit. Berry’s conjecture connects this operator to quantum chaos, providing a physical realization of the audit Hamiltonian. The Montgomery–Odlyzko law and the GUE statistics of the zeros provide overwhelming evidence that such an operator exists, even if its explicit form remains unknown. The SCX framework thus offers not merely a rephrasing of known results but a quantitative methodology for measuring evidence accumulation across independent mathematical approaches. Applied to the Riemann Hypothesis, it confirms that the audit of primes is the cleanest, most thoroughly verified audit in mathematics.

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